Toxoplasmosis Dot-Blot Diagnosis Report

PoCT – Toxoplasmosis Diagnosis Device · anti-GRA7 dot-blot reader

Report date/time	**2026-11-11 09:27:45**	Classifier	**Q4 texture + MLP**
Sample ID	**LAB-0042**	Illumination (model)	**250 lux**
Capture file	**20261111_092739_crop.png**	Classifier build	**2026-11-10T15:41:36**

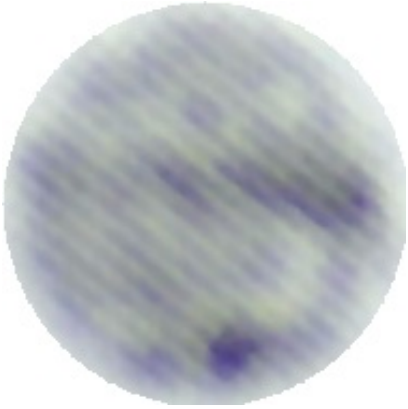
INDETERMINATE

inconclusive – no reliable call

model score 0.099 · calibrated probability ≈ 35 %

Warning: Mean optical density of this spot (0.210) is 4.6× the median of the reference spots (0.046): the spot lies far outside the range the classifier was trained on. Check illumination (≈250 lux), focus, exposure and the sample before relying on this result.

1. Captured spot and acquisition quality



Acquisition QC	**passed**
Detection confidence	**0.96**
Equivalent spot radius	**93.6 px**
Mask area / solidity	**27536 px² / 0.994**
Crop rescaling	**× 0.84 (to training spot size)**
Inner ROI	**85 % of the equivalent radius**
Mean OD in ROI (± SD)	**0.199 ± 0.057**

2. Quantitative decision-support figures

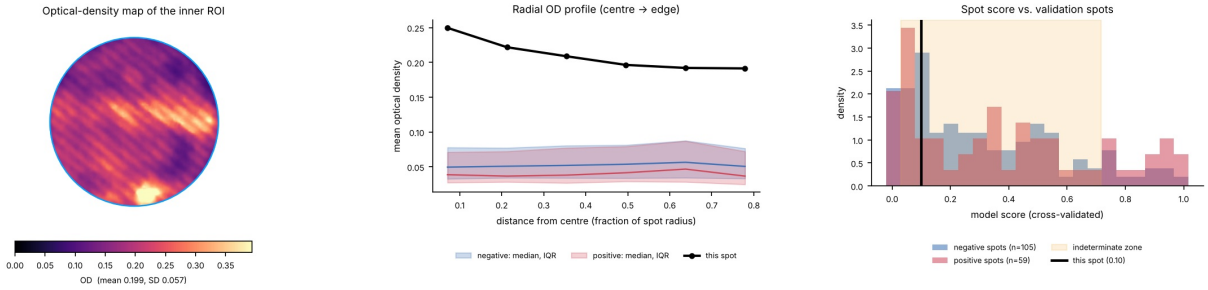


Fig. 1 optical density inside the inner ROI (blue circle), the only region that is quantified; contour size and shape are excluded to avoid pressing/cropping artefacts. **Fig. 2** mean OD in six concentric rings against the median and interquartile range of the negative and positive reference spots. **Fig. 3** score of this spot against the cross-validated scores of the validation spots; the orange band is the indeterminate zone.

3. Descriptive result

- The score 0.10 lies between the NEGATIVE (0.03) and POSITIVE (0.72) thresholds; the classifier cannot make a reliable call for this spot.
- Decision rule: NEGATIVE ≤ 0.03 < INDETERMINATE < 0.72 ≤ POSITIVE (thresholds from cross-validated scores of validation spots; Youden threshold 0.32).
- Radial OD profile (centre → edge): 0.250, 0.222, 0.209, 0.196, 0.192, 0.191; centre-to-edge difference +0.058.

4. Clinical interpretation

- The result is inconclusive: the classifier could not assign the spot to the negative or the positive group with sufficient confidence.

- Re-capture the spot (whole spot in view, in focus, illumination ≈250 lux) and, if possible, test replicate spots of the same serum.
- Use the reference serology (IgM/IgG, IgG avidity) to establish the serostatus; do not base clinical decisions on an inconclusive result.

5. Classifier performance and limitations

Cross-validated metric (single spot)	Value
Spot-level AUC (95 % CI)	0.54 (0.44–0.63)
Serum-level AUC, mean of replicate spots (95 % CI)	0.68 (0.40–0.91)
Label-permutation test	p = 0.15
Share of validation spots with a POSITIVE/NEGATIVE call	24 %
Training data	164 spots, 22 sera (8 positive / 14 negative), membranes 2+3, 250 lux

The 95 % confidence interval of the single-spot AUC includes 0.5: the performance of the classifier on one spot is not distinguishable from chance.

Research prototype. This report was generated by experimental software and is not a validated diagnostic device. It must not be used alone for clinical decisions; confirm with the reference assay and clinical assessment.

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